Jhanavi Putcha

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SUMMARY

AI/ML Engineer with around 4 years of experience in time-series forecasting, reinforcement learning, transformer-based NLP, and generative AI. Skilled in deep learning, LSTM/XGBoost/LightGBM models, real-time streaming analytics, and MLOps pipelines on AWS, GCP, and Snowflake. Expertise in predictive maintenance, anomaly detection, portfolio optimization, and automated report generation. Strong background in feature engineering, model explainability (SHAP/LIME), and CI/CD deployment to drive actionable insights and operational efficiency.

SKILLS

LLM & Generative AI: LLM Fine-Tuning (LoRA, SFT), RAG, BM25 + Dense Retrieval, Hugging Face Transformers, LLaMA, Embeddings, Prompt
Machine Learning & AI: Supervised & Unsupervised Learning, XGBoost, LightGBM, Deep Learning (PyTorch, TensorFlow), CNNs, Transformers, NLP, Time-Series Forecasting, Feature Engineering, Hyperparameter Optimization, A/B Testing, Causal Inference
Cloud & Infrastructure: AWS (S3, EC2, Lambda, SageMaker, RDS, CloudWatch), REST APIs, Microservices Architecture
Engineering, Pinecone, LangChain**,** Multimodal Diffusion Models
Programming & Data: Python (NumPy, Pandas), SQL, PySpark, Spark, Kafka, ETL, Snowflake, Power BI, Tableau
MLOps & Deployment: Docker, Kubernetes, MLflow, CI/CD (GitHub Actions), FastAPI, Model Deployment, Model Monitoring, Drift Detection, Grafana, SHAP Explainability, Model Versioning

EXPERIENCE

Triosoft LLC **May 2025 – Present | New York, Hybrid**
AI/ML Engineer

- Constructed transformer-based NLP models for automated extraction of financial insights from 87+ unstructured reports per week, reducing analyst effort by 18% and improving signal-to-noise ratio in investment research pipelines.
- Engineered reinforcement learning models for portfolio optimization, simulating 24+ market scenarios per asset class, achieving a 6% improvement in risk-adjusted returns and enabling 95+ scenario analyses for senior investment teams.
- Implemented PyTorch/TensorFlow deep learning models for credit risk scoring and market anomaly detection, processing >15 million transaction records monthly, achieving 93% coverage of critical high-risk events.
- Deployed generative AI models and LLMs to automatically generate investment summaries from 80+ financial reports weekly, improving report turnaround time by 20% and enhancing portfolio managers' decision-making efficiency.
- Orchestrated end-to-end MLOps pipelines on AWS for model training, evaluation, deployment, and drift monitoring, supporting 95+ automated model retraining cycles per quarter with <5 sec latency for real-time scoring.
- Fine-tuned large language models using Hugging Face for automated extraction and summarization of 78+ financial reports weekly, improving insight generation efficiency by 18% and enabling faster investment decision-making.
- Implemented a RAG-based pipeline combining vector database retrieval with generative AI to produce context-aware investment insights from 59+ financial documents weekly, improving information accuracy by 15% and accelerating portfolio analysts' decision-making.

LTIMindtree **Jan 2022 – Jul 2024 |Hyderabad, India**
Machine Learning Scientist

- Architected time-series forecasting models on Spark and Snowflake for predicting vehicle component failures from IoT sensor data, improving predictive maintenance accuracy by 18% and reducing unplanned downtime by 12% across 25+ production lines.
- Engineered real-time anomaly detection pipelines using LSTM and XGBoost on sensor and telemetry data streams, identifying 83 critical anomalies per month and increasing system reliability by 15%.
- Designed end-to-end ETL pipelines in Python and PySpark to process 20 TB/month of vehicle data, ensuring low-latency (<5 sec) ingestion and enabling 95% data availability for ML model training.
- Developed interactive dashboards using Power BI, providing 95+ actionable KPIs across fleet health, predictive maintenance, and production efficiency, driving decision-making for 17+ cross-functional stakeholders.
- Optimized hyperparameter tuning and model performance for LightGBM, XGBoost, and LSTM models, improving forecasting precision by 12% and reducing MAE/RMSE by 15% on vehicle sensor datasets.
- Deployed models on AWS SageMaker, integrating MLOps pipelines for CI/CD, automated retraining, and version control, supporting 95+ scheduled model updates per quarter with zero downtime.
- Engineered 25+ features per vehicle from telemetry, sensor, and maintenance data, improving predictive model performance by 18% and covering 95% of critical fleet components.
- Constructed a real-time vehicle telemetry anomaly detection system using Kafka streams and Transformer-based time-series models on AWS SageMaker, detecting 87+ critical anomalies per month and reducing unplanned maintenance by 17%.

PROJECTS

Intelligent Student Performance Predictor with Early Intervention System

- Built predictive models using Random Forest, XGBoost, and LSTM to forecast student academic performance with 85–90% accuracy, leveraging 20+ engineered features from attendance, assignments, and engagement metrics, enabling early identification of at-risk students.
- Developed an interactive Flask-based dashboard with SHAP explainability to visualize predictions and feature impact, allowing educators to understand key performance drivers and intervene 2–3 weeks before critical exams, improving actionable insights and decision-making.

AI-Powered Credit Risk Assessment System

- Engineered a multi-class classification system to categorize students into performance tiers, combining historical grades, participation, and behavioral patterns, achieving 87% prediction accuracy and enabling targeted academic support programs.
- Performed feature selection and dimensionality reduction using PCA and correlation analysis, improving model efficiency and interpretability while maintaining 87% prediction accuracy for credit risk classification.

EDUCATION

Master of Science in Artificial Intelligence **|** University at Buffalo, Buffalo, New York